



Analog IC Design

CMOS Technology

Project 01

TWO-Stage Miller OTA

MOHAMMED NASR EL-DIN

Supervised by Dr. Hesham Omran



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Summary

This report presents Project 01 of the Analog IC Design (CMOS Technology) Summer Training program at the Information Technology Institute (ITI), supervised by Dr. Hesham Omran.

The objective of this project is to design, size, and simulate a Two-Stage Miller-Compensated OTA using the g_m/I_D design methodology in a 180nm CMOS technology, meeting a target specification set including gain, CMRR, phase margin, slew rate, output swing, and power consumption constraints.

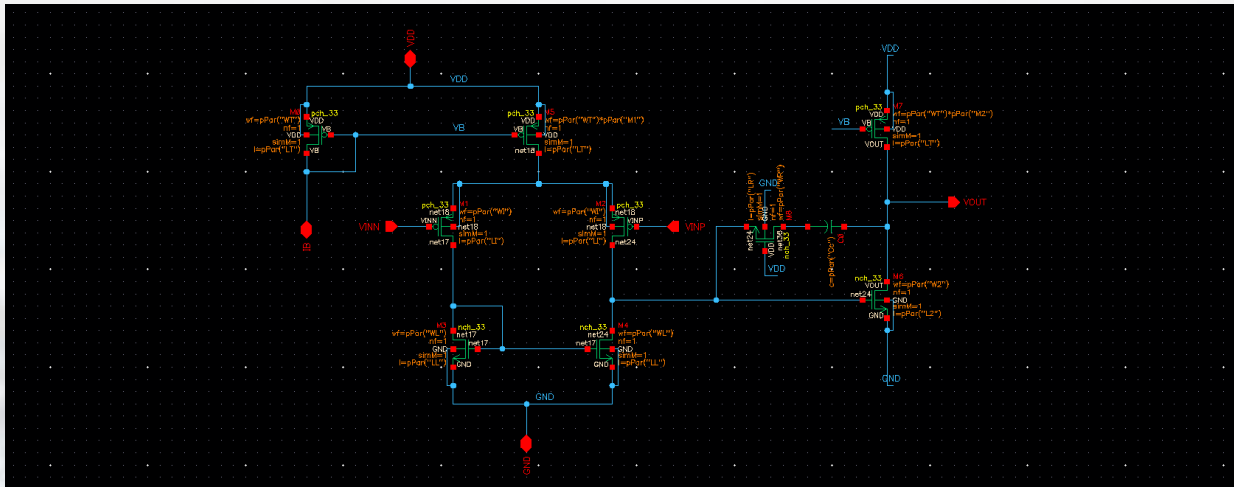
The work covers the full design flow: deriving specs from the required buffer configuration and static gain error, extracting design points for all transistors using g_m/I_D charts, and sizing the input pair, active load, tail current sources, mirror, second-stage input, and the compensation network (including the NMOS voltage-controlled zero-nulling resistor).

The design was verified through open-loop and closed-loop simulations covering DC operating point, AC response, CMRR, stability, transient behavior, and noise, using Cadence Virtuoso for schematic capture and simulation and ADT (Analog Designer's Toolbox) for g_m/I_D based sizing.

Specs and Circuit Topology

Technology	0.13um	0.18um
Supply voltage	1.2V	1.8V
Static gain error	$\leq 0.05\%$	$\leq 0.05\%$
CMRR @ DC	$\geq 74\text{dB}$	$\geq 74\text{dB}$
Phase margin (avoid pole-zero doublets)	$\geq 70^\circ$	$\geq 70^\circ$
OTA current consumption	$\leq 60\mu\text{A}$	$\leq 60\mu\text{A}$
CMIR – high	$\geq 0.6\text{V}$	$\geq 0.8\text{V}$
CMIR – low	$\leq 0.2\text{V}$	$\leq 0.2\text{V}$
Output swing	0.2 – 1V	0.2 – 1.6V
Load	5pF	5pF
Buffer closed loop rise time (10% to 90%)	$\leq 70\text{ns}$	$\leq 70\text{ns}$
Slew rate (SR)	$5\text{V}/\mu\text{s}$	$5\text{V}/\mu\text{s}$

Since the input common mode range is closer to GND rail, the better choice for the 1st stage input is PMOS device. Since the 1st stage DC output is in order of V_{gs3} , it is better to use NMOS device for 2nd stage input as the 1st stage DC output also is in order of V_{gs6} and for good headroom.



Process **TT** at **nominal** temperature.

The technology used: **180nm** $\Rightarrow$ single external supply voltage = **1.8V**

Used an external **10 μA** DC reference current source.

M0: mirror current source

M1, M2: input pair

M3, M4: active load

M5: first stage tail current source I_{B1}

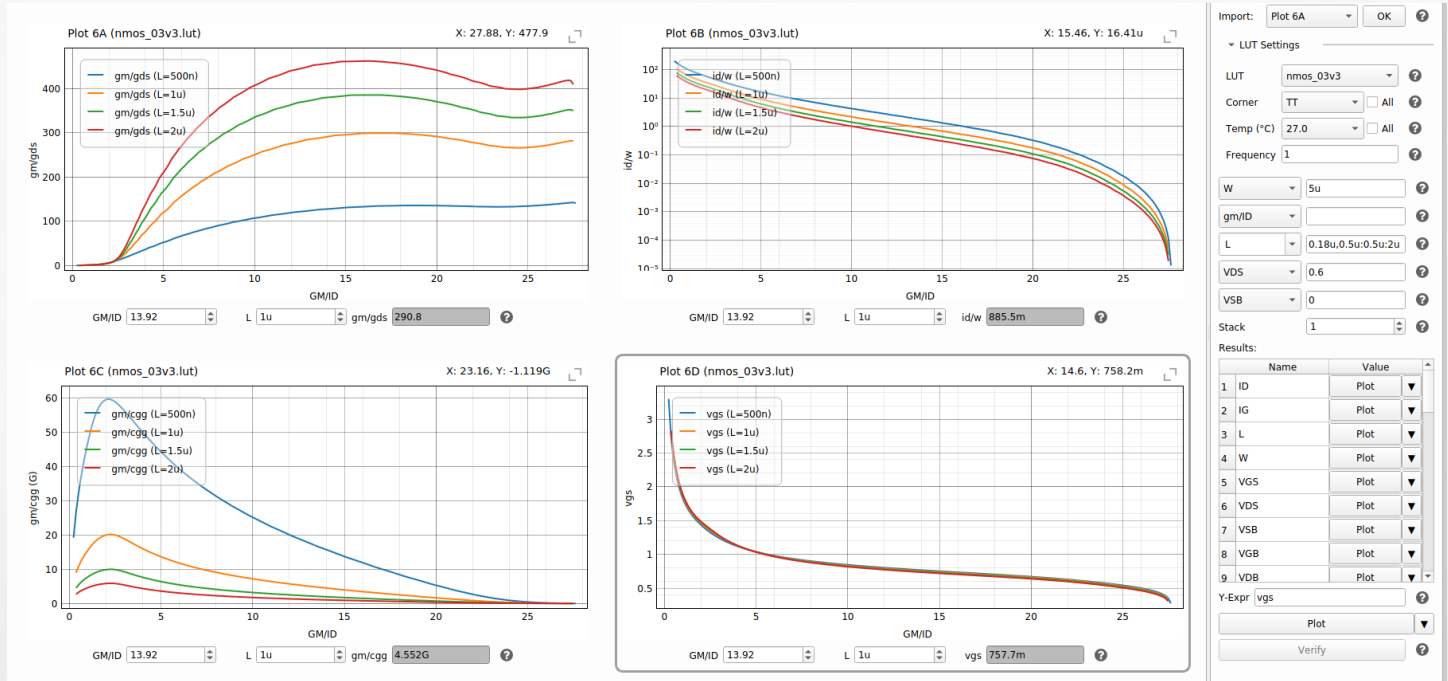
M6: second stage input

M7: second stage tail current source I_{B2}

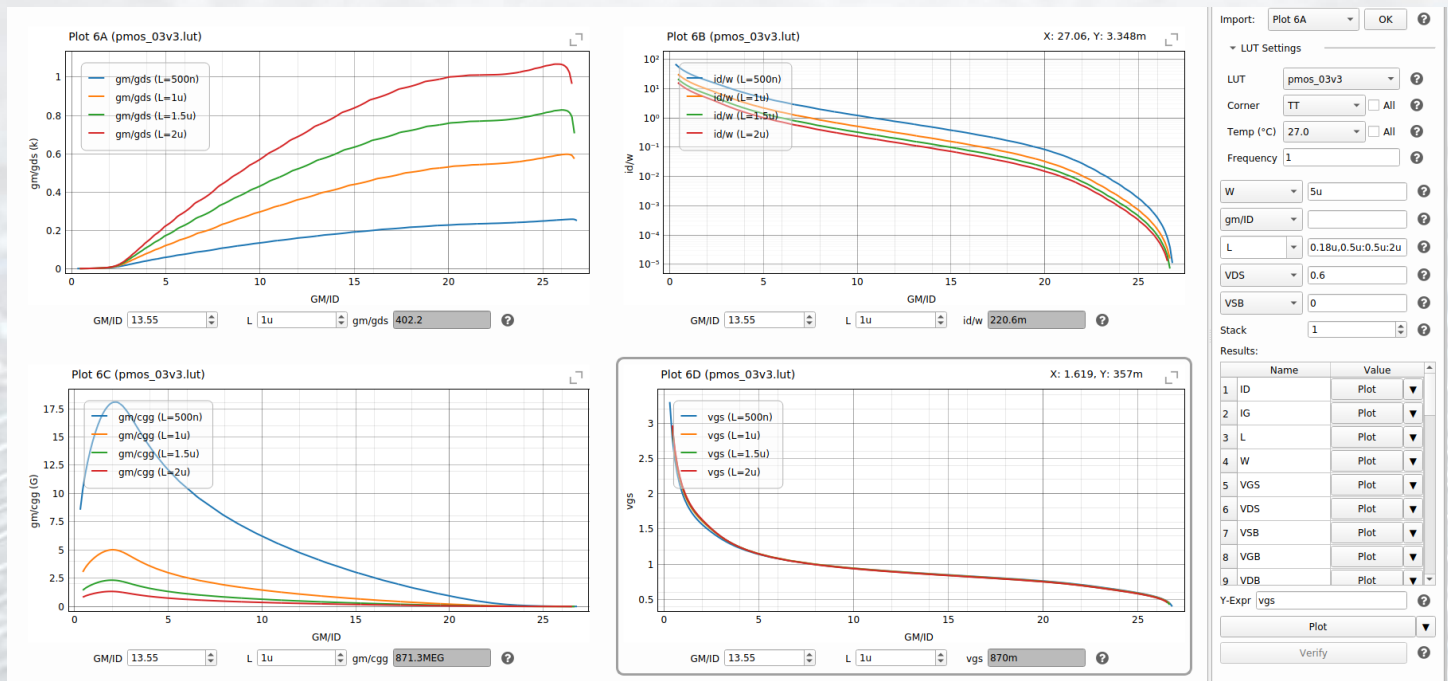
M8: zero-nulling resistor

g_m/I_D Design Charts

Since all of the following figure of merits (FOMs) “ $\frac{g_m}{g_{ds}}$, $\frac{I_D}{W}$, $\frac{g_m}{C_{gg}}$ and V_{GS} ” almost doesn’t depend on the MOSFET width, assume constant width to plot them $W = 5\mu m$



NMOS Charts



PMOS Charts

OTA Design

Since we driving $5pF$ capacitive load, a reasonable starting point for C_c is $2pF < \frac{1}{2}C_L$

$$\text{Slew rate } SR = \frac{I_{B1}}{C_c} = \frac{2I_{D1,2}}{C_c} = 5V/\mu s \Rightarrow I_{D1,2} = 5\mu A, I_{B1} = 10\mu A$$

$$\text{Rise time (from 10\%} \rightarrow 90\%) \approx 2.2\tau \leq 70ns, GBW = UGF \approx \frac{1}{\tau} \geq 31.43 \times 10^6 \equiv 5MHz$$

$$\Rightarrow GBW \approx \frac{g_{m1,2}}{2\pi C_c} \geq 5MHz \Rightarrow g_{m1,2-min} \approx 62.83\mu S$$

$$\text{Let } \frac{g_{m1,2}}{I_{D1,2}} = 15\mu S/\mu A \Rightarrow g_{m1,2} = 75\mu S > 62.83\mu S$$

The closed loop gain $A_{vcl} = \frac{A_{vol}}{1+\beta A_{vol}} = \frac{A_{vol}}{1+A_{vol}}$ for buffer configuration, $A_{vcl} = \frac{A_{vol}}{1+A_{vol}} = \frac{1}{1+\frac{1}{A_{vol}}} \approx 1 - \frac{1}{A_{vol}}$ for large open loop gain $A_{vol} \gg 1$



This graph proof that $\frac{1}{1+x} \approx 1 - x$ for small $x \ll 1$ where $x = \frac{1}{A_{vol}} \Rightarrow A_{vol} \gg 1$

$$\text{Static gain error } \epsilon_s = \frac{1}{LG} = \frac{1}{\beta A_{vol}} \leq 0.05\%, \beta = 1 \Rightarrow A_{vol} \geq \frac{1}{0.05/100} \geq 2000 \equiv 66dB$$

$$A_v = A_{v1} \times A_{v2} \geq 2000, \text{ where } A_{v1} \text{ is the 1}^{st} \text{ stage gain, and } A_{v2} \text{ is the 2}^{nd} \text{ stage gain.}$$

It is better to assign larger gain for the 1st stage, since the 2nd stage noise density is divided by the 1st stage open loop gain referring to the input.

$$\text{Let } A_{v1} = 2A_{v2} \Rightarrow 2A_{v2}^2 \geq 2000 \Rightarrow A_{v2} \geq 31.62 \equiv 30dB, A_{v1} \geq 63.24 \equiv 36dB$$

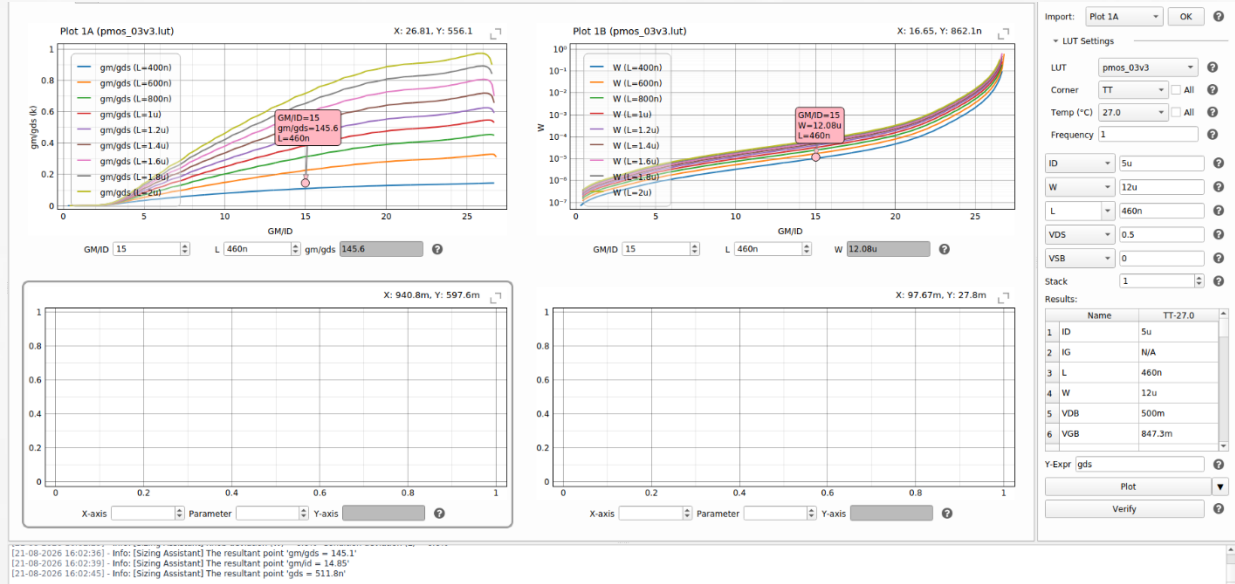
Since $g_{mro} \propto \frac{\sqrt{I_D}}{I_D} = \frac{1}{\sqrt{I_D}}$ at the same sizing, we can set excess gain for the 2nd stage to reduce the power consumed in this stage and consequentially increase efficiency of the overall circuit. And also, to reduce the systematic offset as it will reduce the static gain error.

$$\text{If we take } A_{v1} = 65 \text{ and } A_{v2} = 55 \Rightarrow A_v = 65 \times 55 = 3575$$

PMOS 1st stage input pair $M_{1,2}$

$$V_{SB1} = 0, I_{D1} = 5\mu A, \frac{g_{m1}}{I_{D1}} = 15 \Rightarrow g_{m1} = 75\mu S, A_{v1} \approx g_{m1}(r_{o1}/r_{o3}) = 65$$

$$\text{assume } V_{DS1} = 0.5V, g_{ds1} = g_{ds3} = g_{ds} \Rightarrow \frac{g_{m1}}{2g_{ds}} = 65 \Rightarrow \frac{g_{m1}}{g_{ds}} = 130$$



PMOS 1st stage input pair design point: $L = 460nm, W = 12\mu m, g_{ds} \approx 0.5\mu S, V_{GS} \approx 0.84V$

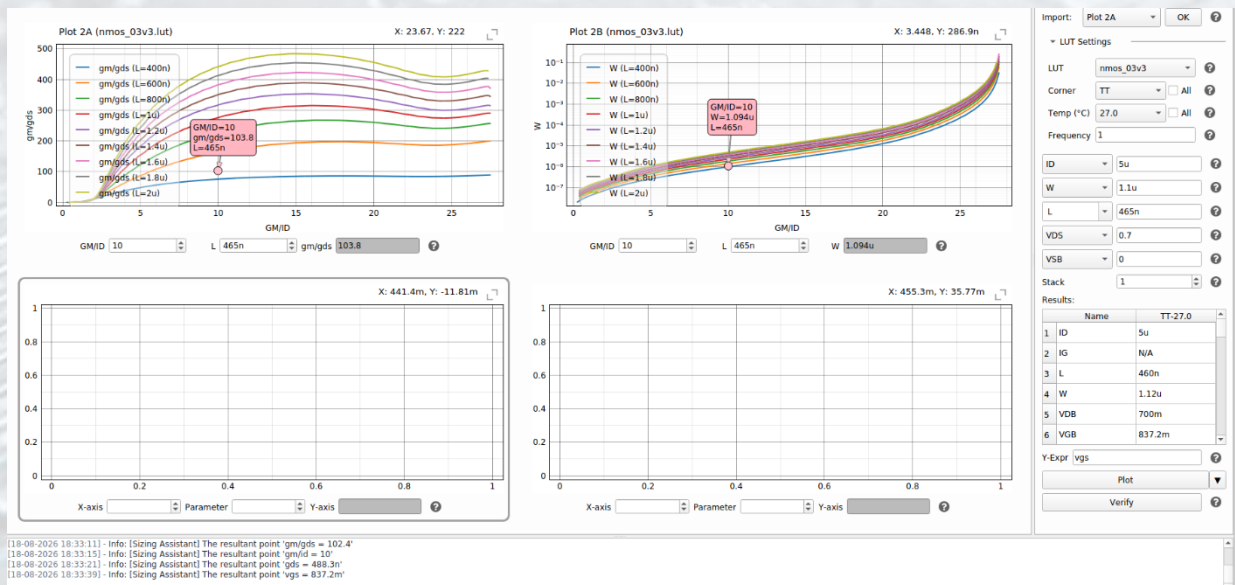
NMOS 1st stage active load $M_{3,4}$

$$V_{SB3} = 0, I_{D1} = 5\mu A, g_{ds} = 0.5\mu S, \text{assume } V_{DS3} = 0.5V$$

As we have to match M_6 with M_3 , $(\frac{g_{m3}}{I_{D1}} = \frac{g_{m6}}{I_{B2}}) \Rightarrow V_3^* = V_6^*$, and to meet the output swing spec ($0.2V \rightarrow 1.6V$),

we should set both V_6^* and V_7^* to be less than or equal to $0.2V \Rightarrow V_3^* \leq 0.2V$, let $\frac{g_{m3}}{I_{D1}} = 10 \Rightarrow g_{m3} = 50\mu S \Rightarrow$

$$\frac{g_{m3}}{g_{ds}} = 100$$



NMOS 1st stage active load design point: $L = 465nm, W = 1.1\mu m, g_{ds} \approx 0.5\mu S$

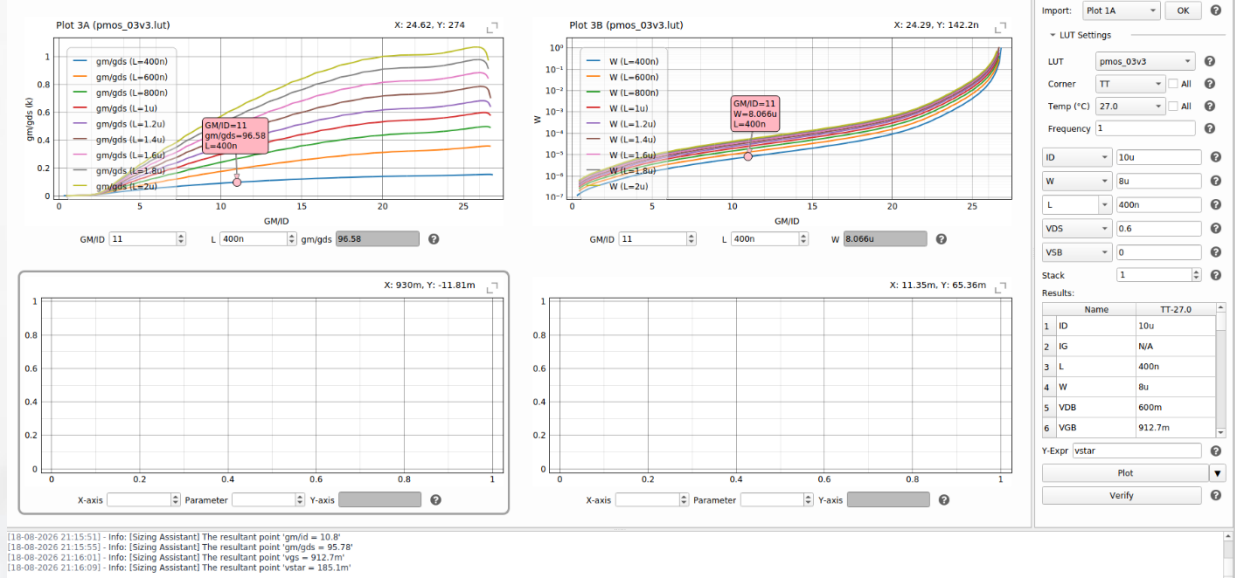
$$CMRR = \frac{A_{vd}}{A_{vcm}} = \frac{A_{v1}A_{v2}}{A_{vcm1}A_{v2}} = \frac{A_{v1}}{A_{vcm1}} \geq 74dB \Rightarrow A_{vcm1} \leq 36dB - 74dB = -38dB.$$

$$A_{vcm1} = \frac{1}{2g_{m3}R_{SS}} \leq -38dB, R_{SS} = \frac{1}{g_{ds5}} \Rightarrow g_{ds5} \leq 2g_{m3} \times 10^{-\frac{38}{20}} \approx 1.26\mu S.$$

PMOS 10 μA 1st stage tail current source M_5

$V_{SB3} = 0, I_{B1} = 10\mu A$, assume $V_{DS5} = 0.6V$, take $g_{ds5} = 1.2\mu S < 1.26\mu S$.

Similar to M3 and M6 relation ($\frac{g_{m5}}{I_{B1}} = \frac{g_{m7}}{I_{B2}}$) $\Rightarrow V_5^* = V_7^*$, to meet the output swing spec ($0.2V \rightarrow 1.6V$) we should set both V_6^* and V_7^* to be less than or equal to $0.2V \Rightarrow V_5^* \leq 0.2V$, let $\frac{g_{m5}}{I_{B1}} = 11 \Rightarrow g_{m5} = 110\mu S \Rightarrow \frac{g_{m5}}{g_{ds5}} \approx 91.67$



PMOS 10 μA tail current source design point: $L = 400nm, W = 8\mu m, V_{GS} \approx 0.9V, V^* < 0.2V$

ω_{pnd}/ω_u	Quality factor Q	ω_u/ω_{u1}	Phase margin PM°
1	1	0.786	51.8
2	0.707	0.910	65.5
3	0.577	0.953	72.4
4	0.500	0.972	76.3
5	0.477	0.981	78.9
6	0.408	0.987	80.7
7	0.378	0.990	81.7
8	0.354	0.992	82.9
9	0.333	0.994	83.7
10	0.316	0.995	84.3
∞	—	1	90

ω_{u1} is the asymptotes UGF, and ω_u is the actual UGF

Choosing $\frac{\omega_{pnd}}{\omega_u} > 3$ gives more than $72.4^\circ PM$, assume $\frac{\omega_{pnd}}{\omega_u} = 3.2 \Rightarrow \frac{G_{m2}}{C_L} = 3.2 \times \frac{G_{m1}}{C_c} \Rightarrow G_{m2} = 8G_{m1}$

Assume $G_m \propto I_D \Rightarrow I_{B2} \approx 8I_{D1} = 4I_{B1} \Rightarrow I_{B2} = 40\mu A$

Total current consumed by the OTA $I_{B1} + I_{B2} = 10\mu A + 40\mu A = 50\mu A < 60\mu A$

PMOS mirror current source M_0

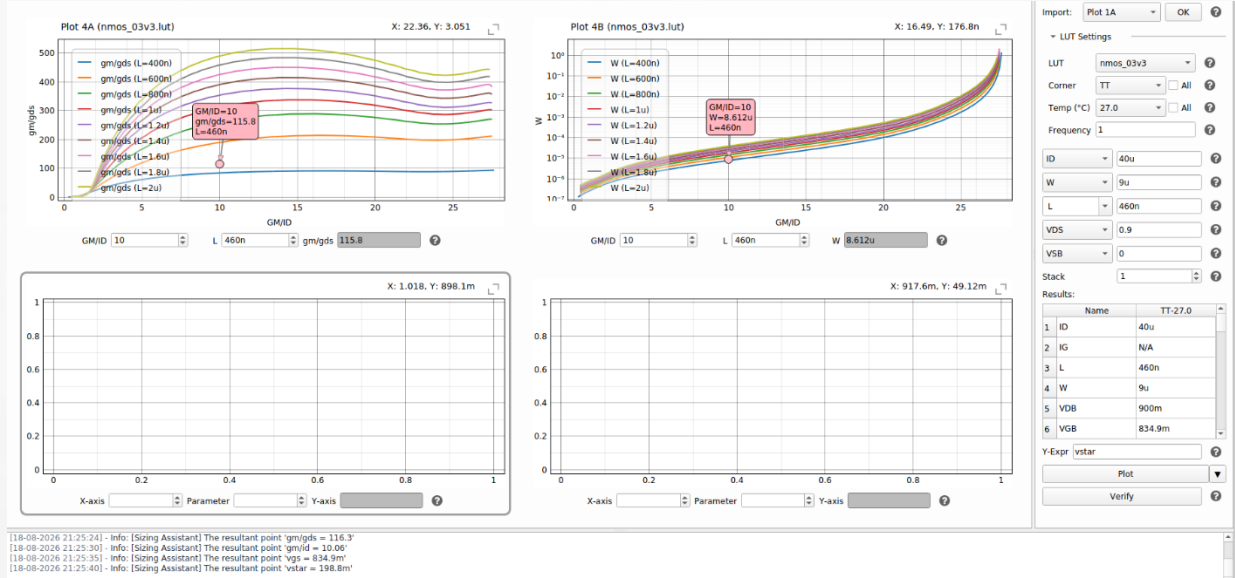
Since the mirroring ratio between M0 and M5 is 1, M0 will have the same sizing of M5 $W_0 = 8\mu A$

NMOS 2nd stage input M_6

$$V_{SB} = 0, I_D = I_{B2} = 40\mu A, A_{v1} \approx g_{m6}(r_{o6}/r_{o7}) = 55$$

Since $V_{GS7} \approx V_{GS5} \approx 0.9V$, assume $V_{DS6} = 0.9V$

$$\text{assume } g_{ds6} = g_{ds7} = g_{ds} \Rightarrow \frac{g_{m6}}{2g_{ds}} = 55 \Rightarrow \frac{g_{m6}}{g_{ds}} = 110$$



NMOS 2nd stage input design point: $L = 460nm, W = 9\mu m, V^* < 0.2V$

PMOS $40\mu A$ 2nd stage tail current source M_7

Since the mirroring ratio between M0 and M7 is 4, M7 will have quadruple the width of M5 $W_7 = 32\mu A$.

NMOS voltage-controlled compensation resistor M_8

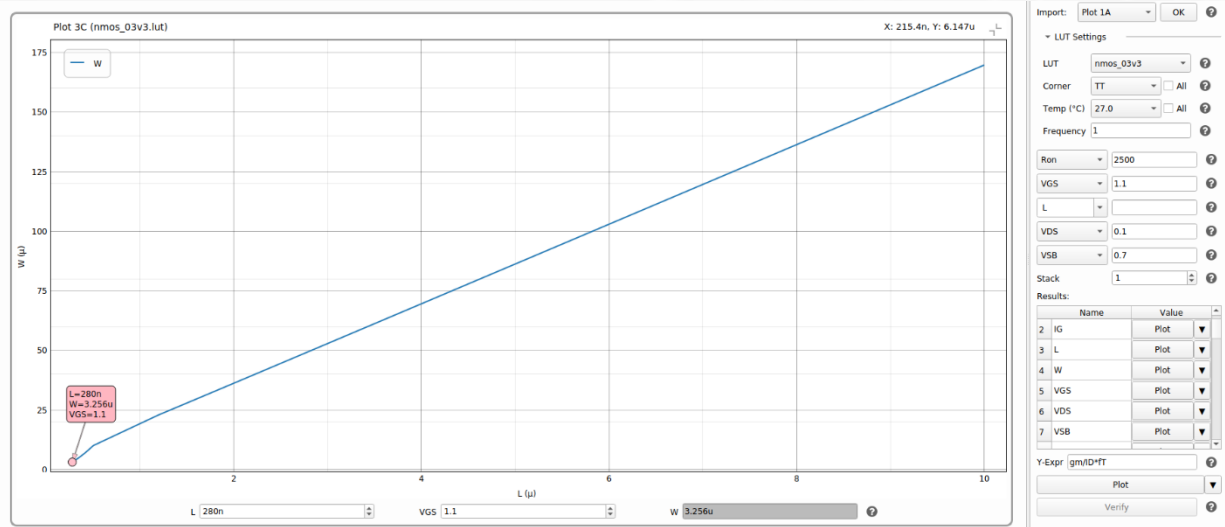
$$\omega_Z = \frac{1}{C_c \left(\frac{1}{g_{m6}} - R_Z \right)}$$

We can save current mirror branch connection, more power consumption, and chip area by connected the gate of M8 with V_{DD} directly without biasing circuit. **Connect V_G with V_{DD}**

Since the source connected to the 1st stage output “not grounded” we can’t connect the body with the source as we using CMOS technology that will make M8 suffers from the body effect. **Connect V_B with GND**

To make sure M8 is in triode, we have to minimize V_{DS8}

To place ω_Z at infinity $R_Z = \frac{1}{g_{m6}} = \frac{1}{400\mu S} = 2500\Omega$, $V_{SB8} = V_{DS3} = 0.7V$, $V_{GS8} = 1.8 - V_{DS3} = 1.1V$, assume $V_{DS8} = 0.1V$



NMOS voltage-controlled resistor design point: $L = 280\text{nm}$, $W = 3.25\mu\text{m}$ “minimized sizing”

MOSFET parameters table

M_X	$W(\mu\text{m})$	$L(\text{nm})$	$g_m(\mu\text{S})$	$I_D(\mu\text{A})$	$\frac{g_m}{I_D}(\mu\text{S}/\mu\text{A})$	$V_{dsat}(\text{mV})$	$V_{ov}(\text{mV})$	$V^*(\text{mV})$
M_0	8	400	108	10	10.8	161.8	135.8	185.1
$M_{1,2}$	12	460	74.86	5	14.85	109.2	63.75	134.7
$M_{3,4}$	1.1	465	50	5	10	168.1	132.4	199.9
M_5	8	400	108	10	10.8	161.8	135.8	185.1
M_6	9	460	402.3	40	10.06	166.7	130.5	198.8
M_7	32	400	432.1	40	10.8	161.8	135.8	185.1
M_8	3.25	280	1.4	0.4	3.5	245.2	256.1	562.5

The above results are getting from ADT

Total area $33.913\mu\text{m}^2$

Note, M8 results are not accurate since we force very small current I_{G6} through M8, and we can get the actual results from the simulator.

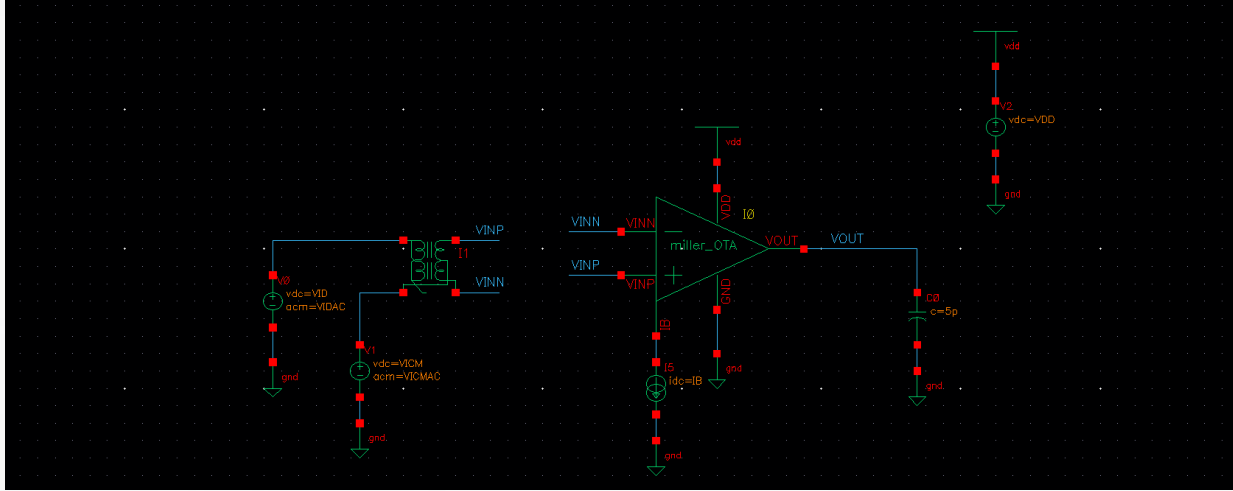
$$V_{ICM} \geq -V_{GS1} + V_1^* + V_{GS3}$$

$$\text{Let } V_{GS3-max} = 0.9\text{V} \Rightarrow V_{ICM} \geq -0.84 + 0.134 + 0.9 = 0.194\text{V} = V_{ICM-min}$$

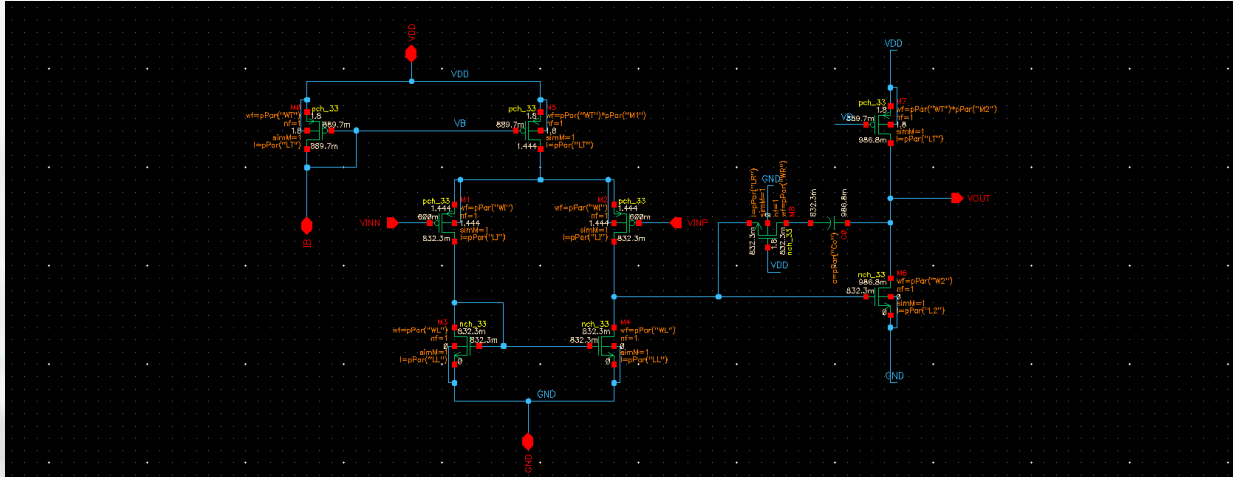
$$V_{ICM} \leq V_{DD} - V_{dsat5} - V_{GS1} = 1.8 - 0.16 - 0.84 = 0.8\text{V} = V_{ICM-max}$$

$$0.194\text{V} \leq V_{ICM} \leq 0.8\text{V} \Rightarrow CMIR_{low} < 0.2\text{V}, CMIR_{high} = 0.8\text{V} \text{ that meets the output swing spec.}$$

Open-Loop OTA Simulation



DC OP analysis



Open loop OTA and bias circuit: DC voltage annotated, Setting $V_{ICM} = \frac{V_{DD}}{3} = 0.6V$

Output	Nominal
M0 id	-10u
M0 gm	110.7u
M0 region	2
M1 id	-4.692u
M1 gm	71.74u
M1 region	2
M2 id	-4.692u
M2 gm	71.74u
M2 region	2
M3 id	4.692u
M3 gm	47.65u
M3 region	2
M4 id	4.692u
M4 gm	47.65u
M4 region	2
M5 id	-9.385u
M5 gm	104.4u
M5 region	2
M6 id	39.44u
M6 gm	403.7u
M6 region	2
M7 id	-39.44u
M7 gm	438.3u
M7 region	2
M8 id	-832.3f
M8 gm	4.437p
M8 region	1

The results are almost as expected.

For M8, $I_D = I_{G6} \approx 835fA$, $g_m \approx 4.5pS \Rightarrow \frac{g_m}{I_D} \approx 5.39$, but we it keeps in triode as V_{DS} is minimized.

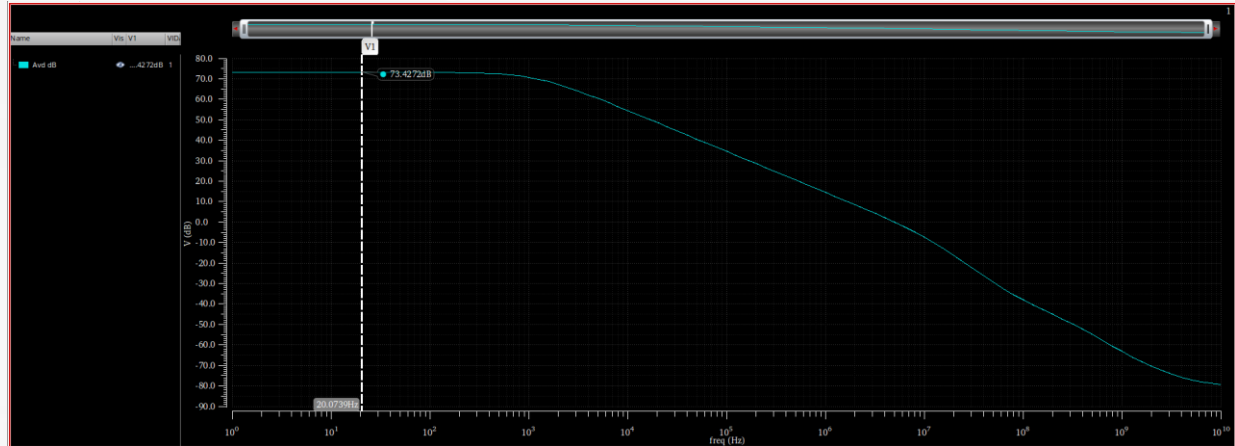
Note that:

The current and g_m for the input pair are exactly equal.

1st stage DC output V_{GS6} follows the mirror node $V_{GS3} = 0.832V$.

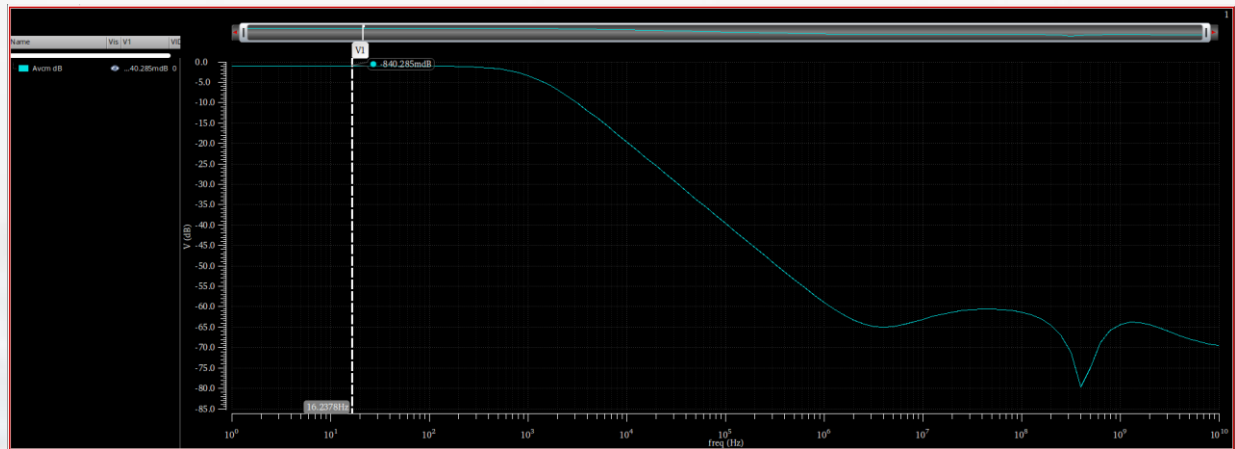
The 2nd stage DC output is $V_{DS6} = 0.986V$.

Diff small signal



$A_{vd}(dB)$

CM small signal



$A_{vcm}(dB)$, DC value = $-0.84dB$

Hand calculations

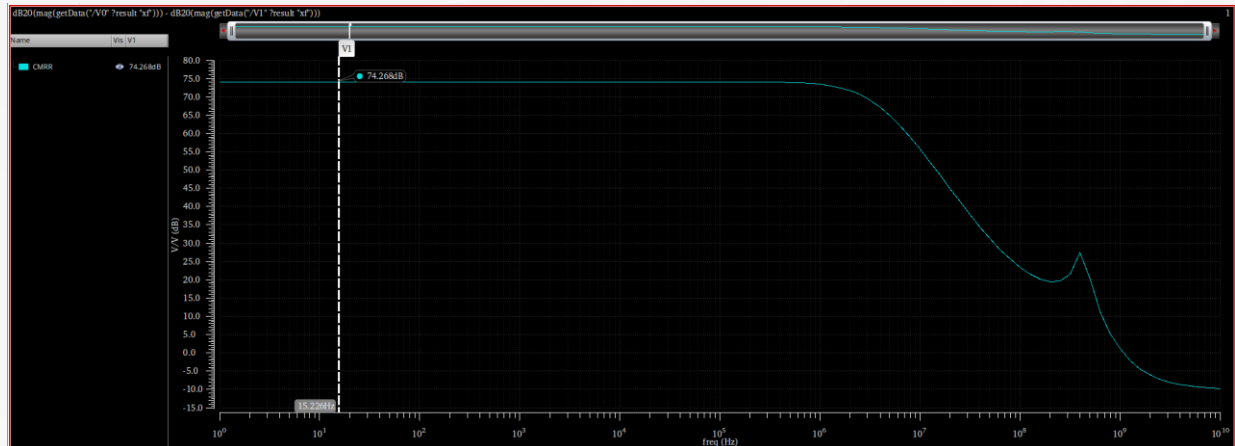
$$A_{vd} = A_{v1} \times A_{v2} \geq 65 \times 50 = 3250 \equiv 70.24dB, GBW \approx \frac{g_{m1}}{2\pi C_c} = \frac{75\mu S}{2\pi \cdot 2pF} \approx 5.97MHz \Rightarrow BW = \frac{GBW}{A_{vd}} \approx 1.8KHz$$

$$A_{vcm} = \frac{g_{ds5}}{2g_{m3}} \times A_{v2} = \frac{1.26\mu S}{2 \times 50\mu S} \times 55 = 0.693 \equiv -3.18dB \text{ note the common mood gain increases as we increased the 2}^{nd} \text{ stage gain.}$$

Parameter	Simulation	Hand calculations
DC diff gain	73.43dB	70.24dB
BW	1.18KHz	1.8KHz
GBW	5.52MHz	5.97MHz
DC CM gain	-0.84dB	-3.18dB

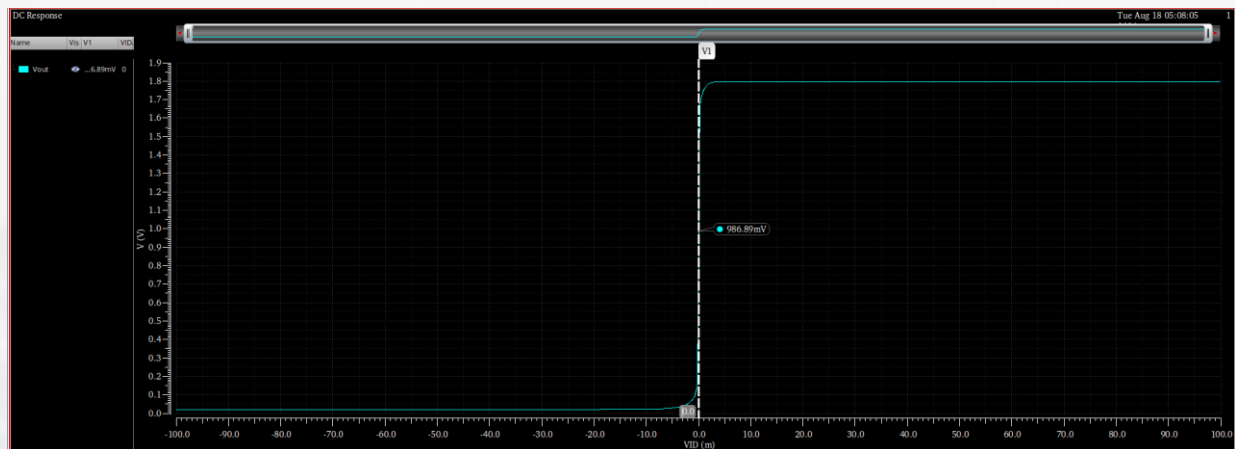
Output	Nominal
Avd	4.693k
Avd dB	73.43
BW	1.175k
UGF	5.164M
GBW	5.515M

CMRR

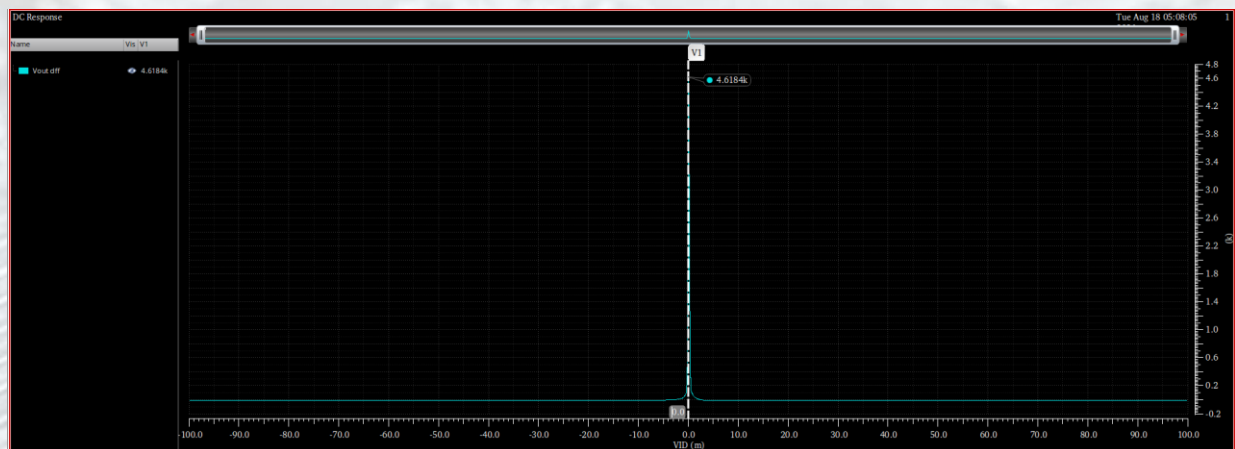


CMRR (dB) 74.3dB

Diff large signal



V_{out} vs V_{id} , $V_{out} = 0.987V$ at $V_{id} = 0$ same as the DC OP analysis results



V_{out} derivative vs V_{id} , peak value $\approx 4.62K$

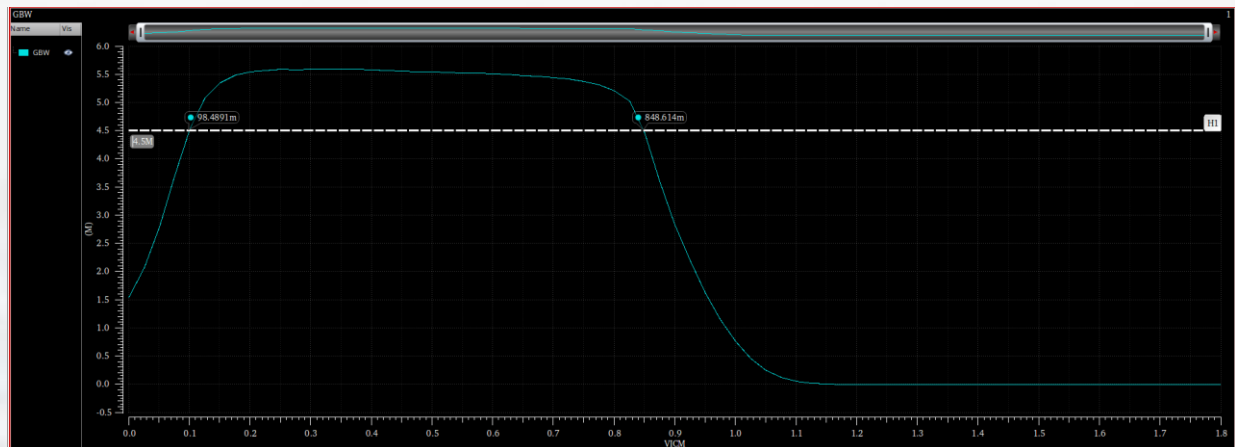
Note: the peak “diff gain” slightly smaller than the AC analysis value, since the accuracy in the large signal simulation is limited by the transient analysis step

CM large signal (*region vs V_{ICM}*)



Region vs V_{ICM} , the CMIR form 0.1V $\rightarrow$ 0.8V

CM large signal (*GBW vs V_{ICM}*)



GBW vs V_{ICM} , the CMIR is from 0.1 $\rightarrow$ 0.85V for 10% error from maximum GBW

Note that, if we are using NMOS input pair, body effect may cause CMIR to extend till VDD, and If we are using PMOS input pair, body effect may cause CMIR to extend till GND.

In general, the body effect increases the magnitude of the threshold voltage V_{th}

Incase NMOS:

$$V_{th1} + V_{ov1} + V_{DS5} \leq V_{ICM} \leq V_{DD} - |V_{GS3}| + V_{th1}$$

The body effect will increase V_{th1} that will increase both $V_{ICM-max}$ and $V_{ICM-min}$

$\Rightarrow$ CMIR shifts toward V_{DD} rail

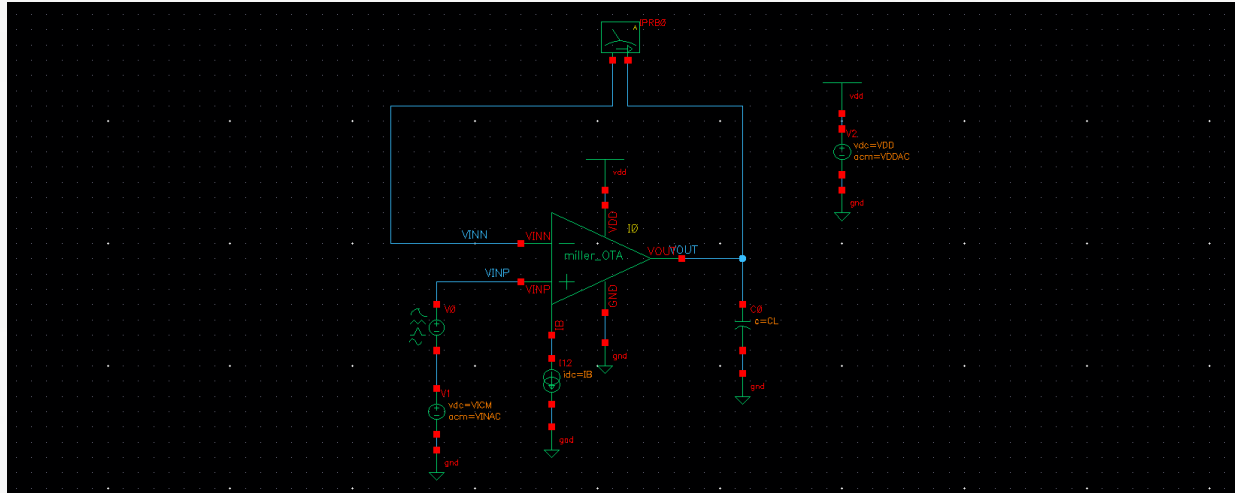
Incase PMOS:

$$-|V_{th1}| + V_{GS3} \leq V_{ICM} \leq V_{DD} - |V_{DS5}| - |V_{th1}| + |V_{ov1}|$$

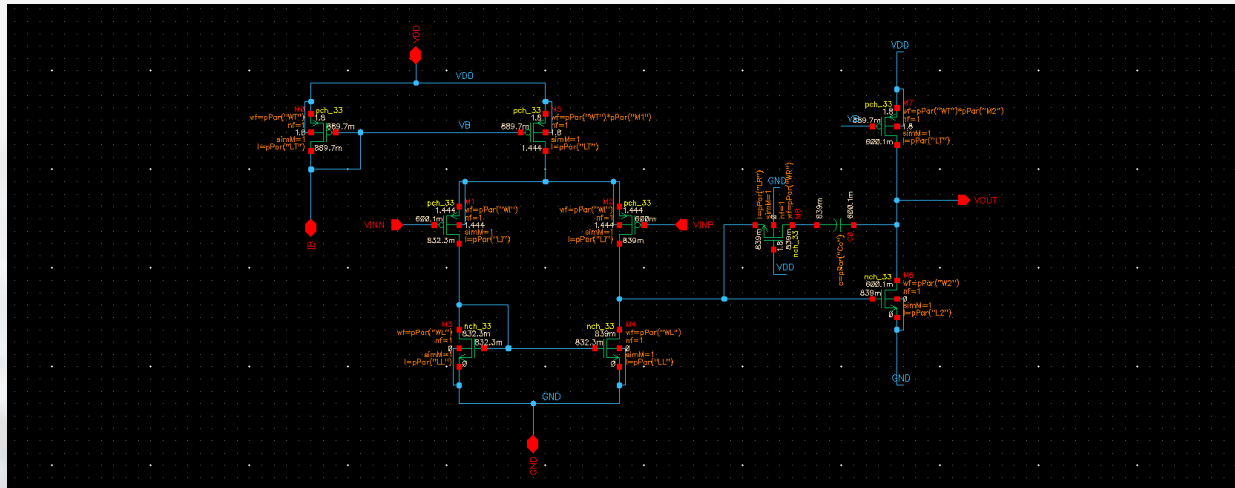
The body effect will increase $|V_{th1}|$ that will decrease both $V_{ICM-max}$ and $V_{ICM-min}$

$\Rightarrow$ CMIR shifts toward GND rail

Closed-Loop OTA Simulation



DC OP analysis



Closed loop OTA and bias circuit: DC voltage annotated Setting $V_{ICM} = \frac{V_{DD}}{3} = 0.6V$

Output	Nominal
M0 id	-10u
M0 gm	110.7u
M0 region	2
M1 id	-4.691u
M1 gm	71.73u
M1 region	2
M2 id	-4.694u
M2 gm	71.75u
M2 region	2
M3 id	4.691u
M3 gm	47.65u
M3 region	2
M4 id	4.694u
M4 gm	47.67u
M4 region	2
M5 id	-9.385u
M5 gm	104.4u
M5 region	2
M6 id	40.78u
M6 gm	409.2u
M6 region	2
M7 id	-40.78u
M7 gm	450.7u
M7 region	2
M8 id	-841.4f
M8 gm	4.656p
M8 region	1

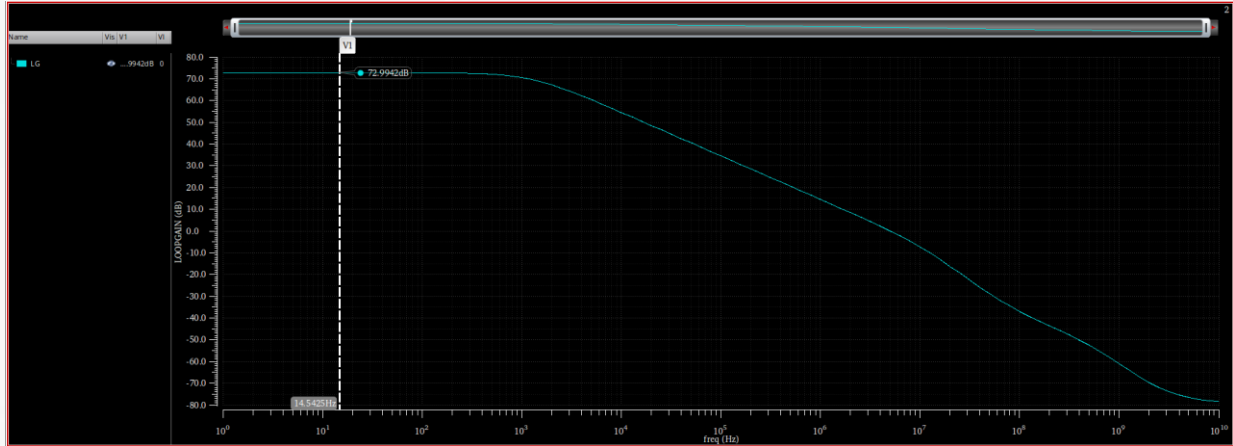
For M8, $I_D = I_{G6} \approx 841.4fA$, $g_m \approx 4.656pS \Rightarrow \frac{g_m}{I_D} \approx 5.53$, but we it keeps in triode as V_{DS} is minimized.

Note that:

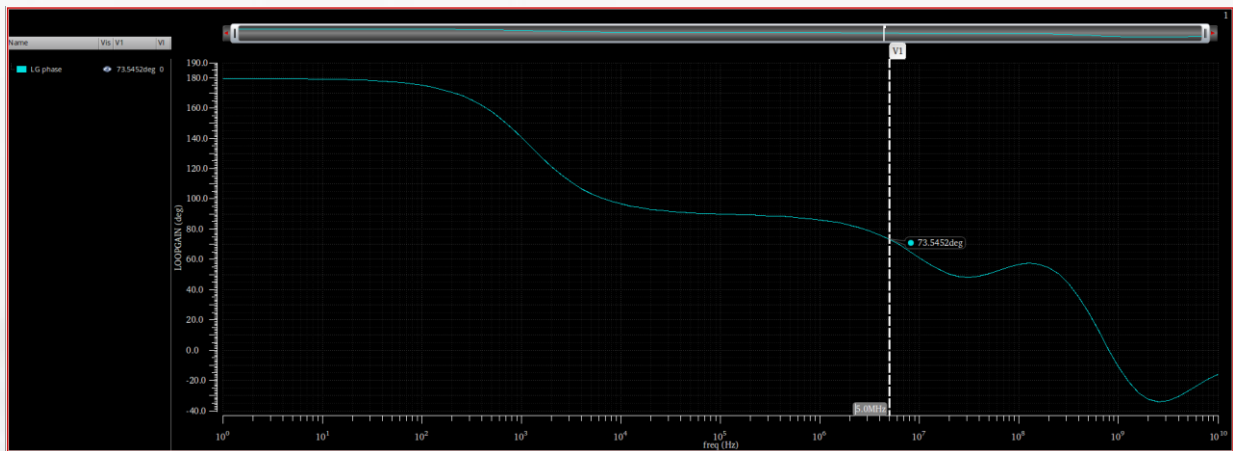
The current and g_m for the input pair are not exactly equal, since V_{INN} follows the output, and the open loop gain is finite. So V_{out} can't exactly equals V_{INP} , that what makes systematic offset " $V_{INN} \neq V_{INP}$ exactly".

$V_{out} = V_{INN} = 600.1mV$, $V_{INP} = 600mV$.

Loop gain simulation



$$LG = 1 \times A_{OL} \text{ (dB)}$$



LG phase

Parameter	Open loop simulation	LG simulation
DC gain	73.43dB	73dB
UGF	5.164MHz	5.174MHz
GBW	5.52MHz	5.514MHz

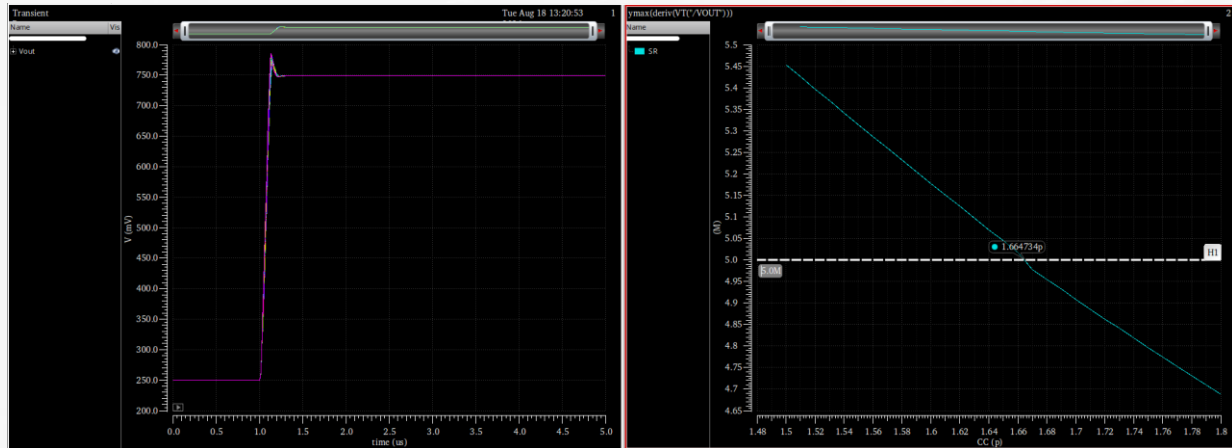
Output	Nominal
DCLG	4.464k
LG dB	72.99
LG BW	1.232k
GBW	5.514M
UGF	5.174M
PM	73.12

Let $\omega_{pnd} = A\omega_u \Rightarrow \frac{G_{m2}}{C_L} = A \frac{G_{m1}}{C_c} \Rightarrow A = \frac{C_c}{C_L} \times \frac{G_{m2}}{G_{m1}}$, let $G_m \propto I_D \Rightarrow A \approx \frac{2}{5} \times \frac{I_{B2}}{I_{D1}} = \frac{2}{5} \times \frac{40}{5} = 3.2 \Rightarrow PM \approx 73^\circ$

Note that, the PM is below critical PM “76.3°” that will show small **ringing in the time** domain response.

Parameter	LG simulation	Hand calculations
DC gain	73dB	73.43dB
GBW	5.514MHz	5.97MHz
PM	73.12°	73°

Slew rate



V_{out} step response

SR vs C_c

Note that, the above graph (SR vs C_c) show that the maximum value for C_c that can meet the SR spec is $1.665pF$.

Output	Nominal
$C_c = 2pF$	
Vout	
Vsig	
SR	4.295M
$C_c = 1.665pF$	
Vout	
Vsig	
SR	5.005M

Slew rate for
 $C_c = 2pF, 1.665pF$

Output	Nominal
DCLG	1.105k
LG dB	60.87
LG BW	5.88k
GBW	6.515M
UGF	6.003M
PM	70.64

Stability simulation
results for $C_c = 1.665pF$
 $PM > 70^\circ$



V_{in} $V_{out-C_c=2pF}$ $V_{out-C_c=1.665pF}$

Hand calculations

$$SR = \frac{I_{B1}}{C_c}$$

$$\Rightarrow SR = 5V/\mu s \text{ for } C_c = 2pF$$

$$\Rightarrow SR = 6V/\mu s \text{ for } C_c = 1.665pF$$

Settling time



■ V_{in} ■ $V_{out-C_c=2pF}$ ■ $V_{out-C_c=1.665pF}$

Note that, the response shows some ringing that inversely proportional to C_c , and this is because the phase margin is below the critical $PM = 76.3$, So the curves will show under damped response “ringing”.

Hand calculations

Rise times $\approx 2.2\tau = \frac{2.2}{GBW} = 2.2 \times \frac{C_c}{g_{m1}}$

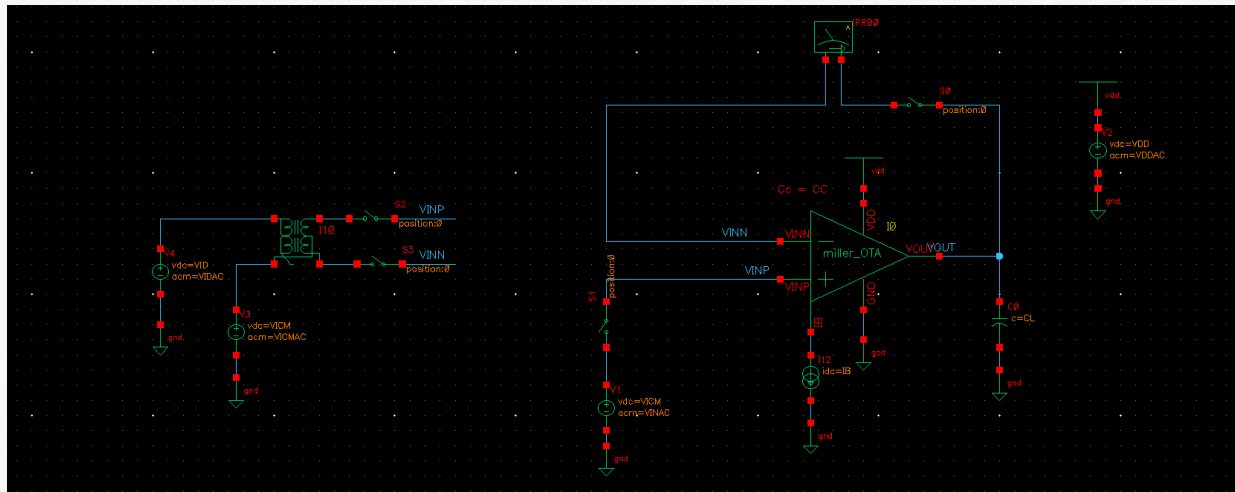
For $C_c = 2pF \Rightarrow T_{rise} = 58.66ns$, For $C_c = 1.665pF \Rightarrow T_{rise} = 48.84ns$

Since $T_{rise} \approx 2.2\tau$ is based on 1st order system, in this 2nd order system will show better results.

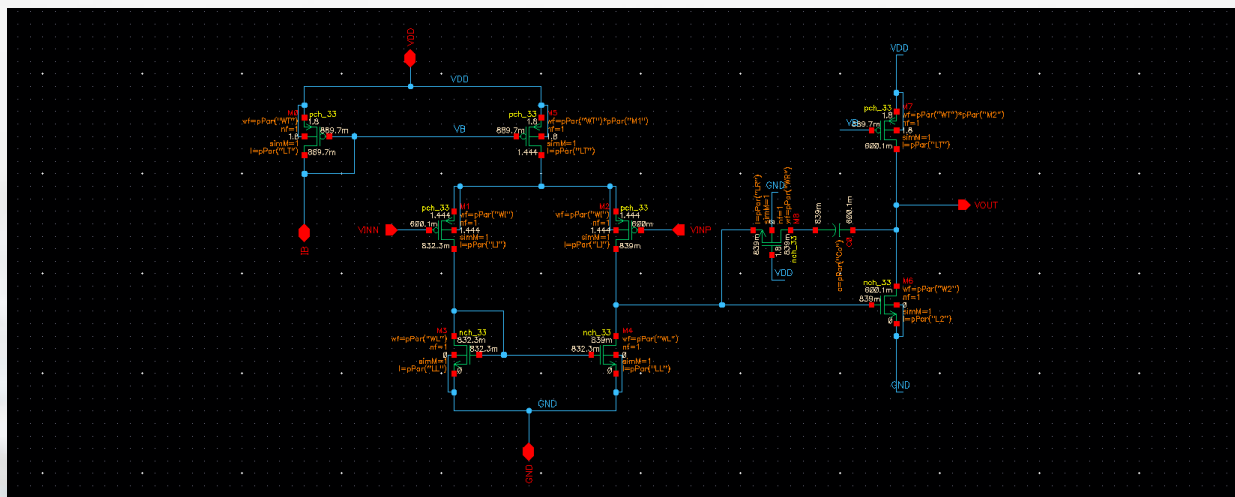
Output	Nominal
$C_c = 2pF$	
Vout	
Vsig	
risetime	46.65n
$C_c = 1.665pF$	
Vout	
Vsig	
risetime	38.93n

Parameter	Simulation		Hand Calculation	
	$C_c = 2pF$	$C_c = 1.665pF$	$C_c = 2pF$	$C_c = 1.665pF$
SR	4.3V/μs	5V/μs	5V/μs	6V/μs
T_{rise}	46.65ns	38.93ns	58.66ns	48.84ns

DC Closed Loop AC Open-Loop OTA Simulation



DC OP analysis



DC Closed loop AC Open loop OTA and bias circuit: DC voltage annotated Setting $V_{ICM} = \frac{V_{DD}}{3} = 0.6V$

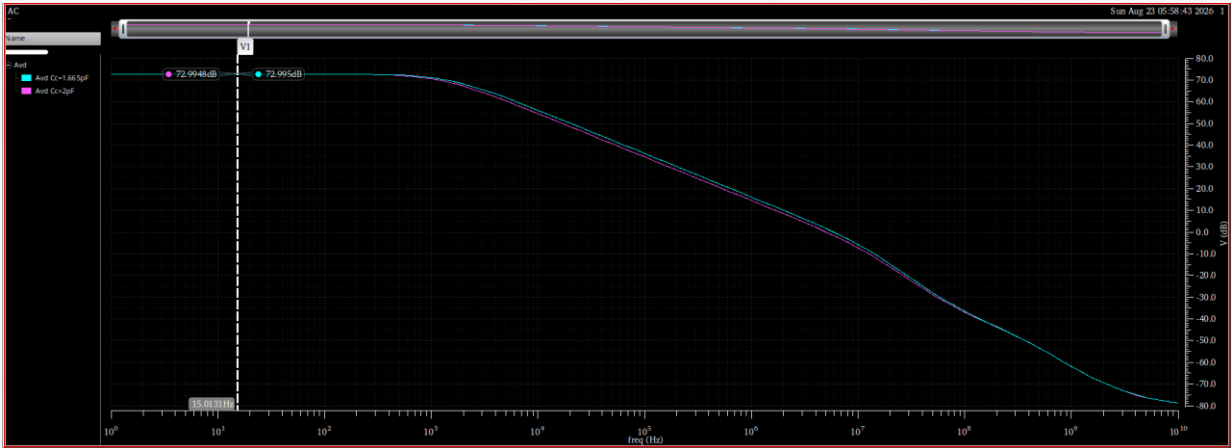
Output	Nominal
M0 id	-10u
M0 gm	110.7u
M0 region	2
M1 id	-4.691u
M1 gm	71.73u
M1 region	2
M2 id	-4.694u
M2 gm	71.75u
M2 region	2
M3 id	4.691u
M3 gm	47.65u
M3 region	2
M4 id	4.694u
M4 gm	47.67u
M4 region	2
M5 id	-9.385u
M5 gm	104.4u
M5 region	2
M6 id	40.78u
M6 gm	409.2u
M6 region	2
M7 id	-40.78u
M7 gm	450.7u
M7 region	2
M8 id	-841.4f
M8 gm	4.656p
M8 region	1

For M8, $I_D = I_{G6} \approx 841.4 \mu A$, $g_m \approx 4.656 pS \Rightarrow \frac{g_m}{I_D} \approx 5.53$, but we it keeps in triode as V_{DS} is minimized.

Note that:

The DC voltages at the input terminals of the op-amp are not exactly equal, Since V_{INN} follows the output in DC value. and the open loop gain is finite, V_{out} can't exactly equals the DC value of V_{INP} , and that what makes systematic offset " $V_{INN} \neq V_{INP}$ "

Diff small signal



A_{vd} ■ $C_c = 2pF$ ■ $C_c = 1.665pF$

Parameter	Open loop simulation		AC Open loop simulation	
	$C_c = 2pF$	$C_c = 1.665pF$	$C_c = 2pF$	$C_c = 1.665pF$
DC diff gain	73.43dB	73.43dB	73dB	73dB
BW	1.18KHz	1.41KHz	1.23KHz	1.48KHz
GBW	5.52MHz	6.62MHz	5.52MHz	6.63MHz

Output	Nominal
$C_c = 2pF$	
Vout	4.465k
Avd	73
Avd dB	1.232k
BW	5.515M
GBW	
$C_c = 1.665pF$	
Vout	4.465k
Avd	73
Avd dB	1.481k
BW	6.627M
GBW	

Note, as C_c decreases the BW and the GBW increase, but the DC gain remain unchanged.
In AC open loop simulation, have slightly larger BW and slightly smaller DC gain but the band width almost same in the two simulations.

Noise simulation

Window Expressions Info Help

cadence

Device	Param	Noise Contribution	% Of Total
I0.M4.m0	fn	6.27893e-09	27.83
I0.M3.m0	fn	6.04648e-09	26.80
I0.M0.m0	id	2.28355e-09	10.12
I0.M1.m0	id	1.55498e-09	6.89
I0.M2.m0	id	1.5468e-09	6.86
I0.M8.m0	id	1.26646e-09	5.61
I0.M4.m0	id	8.99096e-10	3.98
I0.M3.m0	id	8.65875e-10	3.84
I0.M7.m0	id	6.0675e-10	2.69
I0.M6.m0	id	5.64874e-10	2.50
I0.M6.m0	fn	2.39099e-10	1.06
I0.M1.m0	fn	1.93589e-10	0.86
I0.M2.m0	fn	1.93569e-10	0.86
I0.M0.m0	fn	1.67737e-11	0.07
I0.M7.m0	fn	4.51851e-12	0.02
I0.M5.m0	id	1.03906e-12	0.00
I0.M5.m0	fn	7.21414e-14	0.00
I0.M8.m0	fn	2.62689e-25	0.00

Integrated Noise Summary (in V^2) Sorted By Noise Contributors

Total Summarized Noise = 2.25625e-08

Total Input Referred Noise = 0.000193706

The above noise summary info is for noise data with CC = 2e-12

Noise summary for $C_c = 2pF$

Window Expressions Info Help cadence

Device	Param	Noise Contribution	% Of Total
I0.M4.m0	fn	6.40959e-09	26.96
I0.M3.m0	fn	6.17224e-09	25.96
I0.M0.m0	id	2.28785e-09	9.62
I0.M1.m0	id	1.86243e-09	7.83
I0.M2.m0	id	1.85247e-09	7.79
I0.M8.m0	id	1.26454e-09	5.32
I0.M4.m0	id	1.07683e-09	4.53
I0.M3.m0	id	1.03713e-09	4.36
I0.M7.m0	id	6.09319e-10	2.56
I0.M6.m0	id	5.67266e-10	2.39
I0.M6.m0	fn	2.26638e-10	0.95
I0.M1.m0	fn	1.94556e-10	0.82
I0.M2.m0	fn	1.94534e-10	0.82
I0.M0.m0	fn	1.56072e-11	0.07
I0.M7.m0	fn	4.21789e-12	0.02
I0.M5.m0	id	1.14661e-12	0.00
I0.M5.m0	fn	7.27473e-14	0.00
I0.M8.m0	fn	2.47537e-25	0.00

Integrated Noise Summary (in V^2) Sorted By Noise Contributors

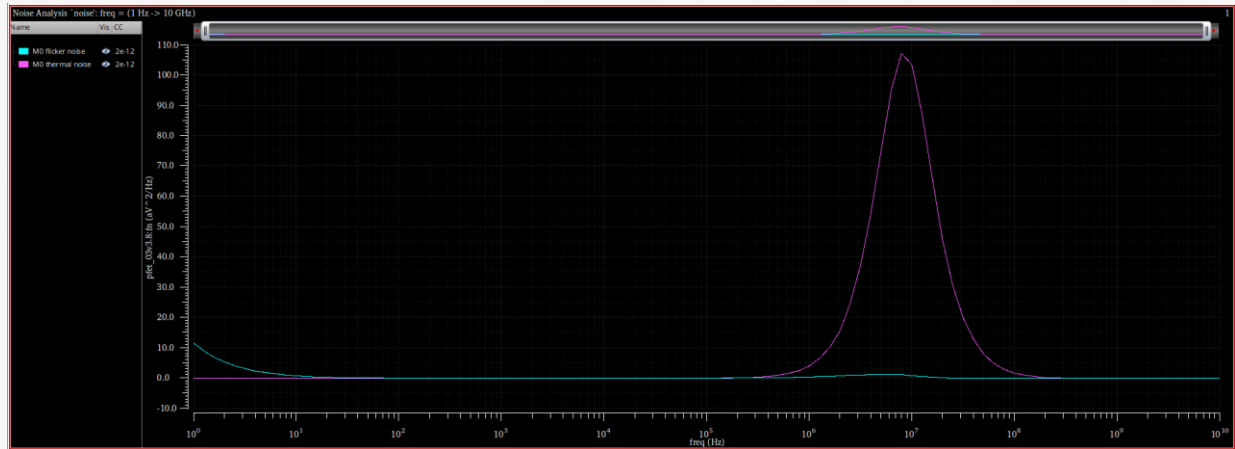
Total Summarized Noise = 2.37764e-08

Total Input Referred Noise = 0.00020222

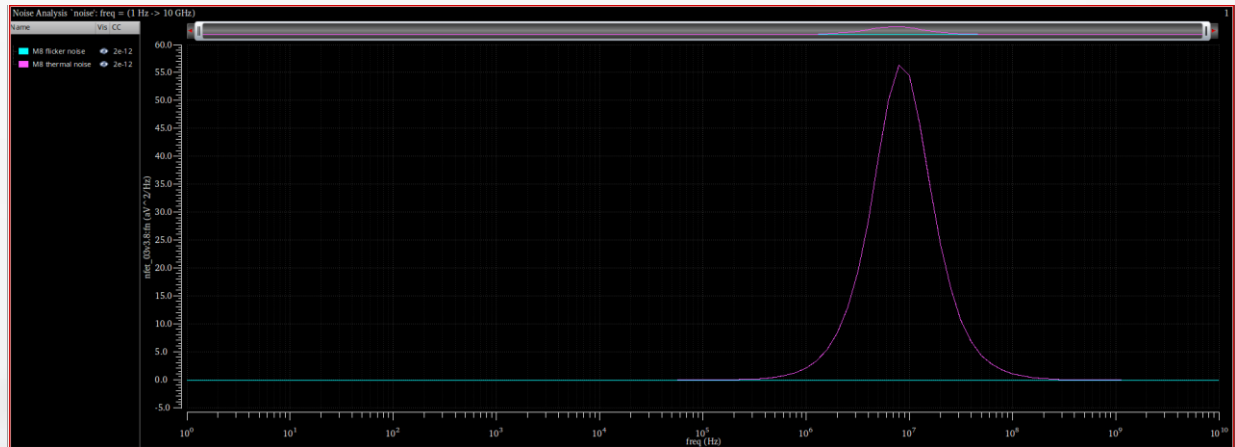
The above noise summary info is for noise data with CC = 1.665e-12

Noise summary for $C_c = 1.665pF$

The results of the noise summary almost are the same for the two capacitance values; we can apply the noise analysis on at any value of them. take $C_c = 2pF$

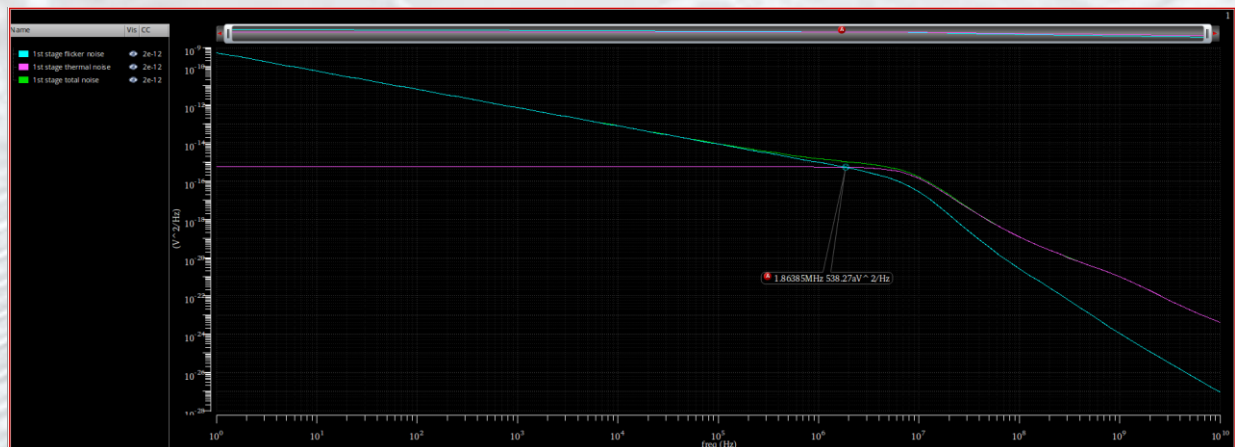


M_0 noise: ■ flicker noise ■ thermal noise

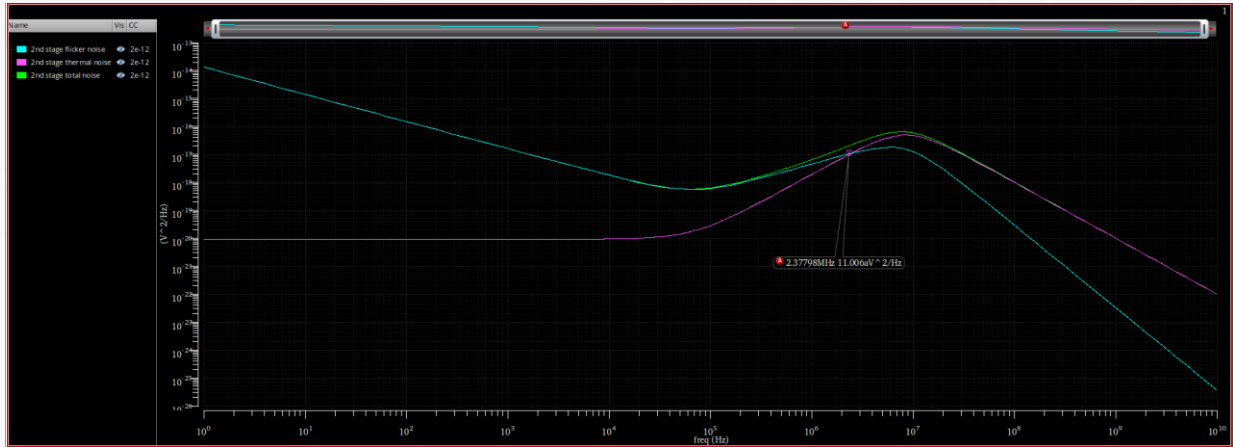


M_8 noise: ■ flicker noise ■ thermal noise

The graphs above show that, M_0 and M_8 noises can almost be neglected at low frequency, but at high frequency, they will significantly contribute in the circuit noise due to the degradation in the input and load MOSFETs because of the parasitic capacitance.

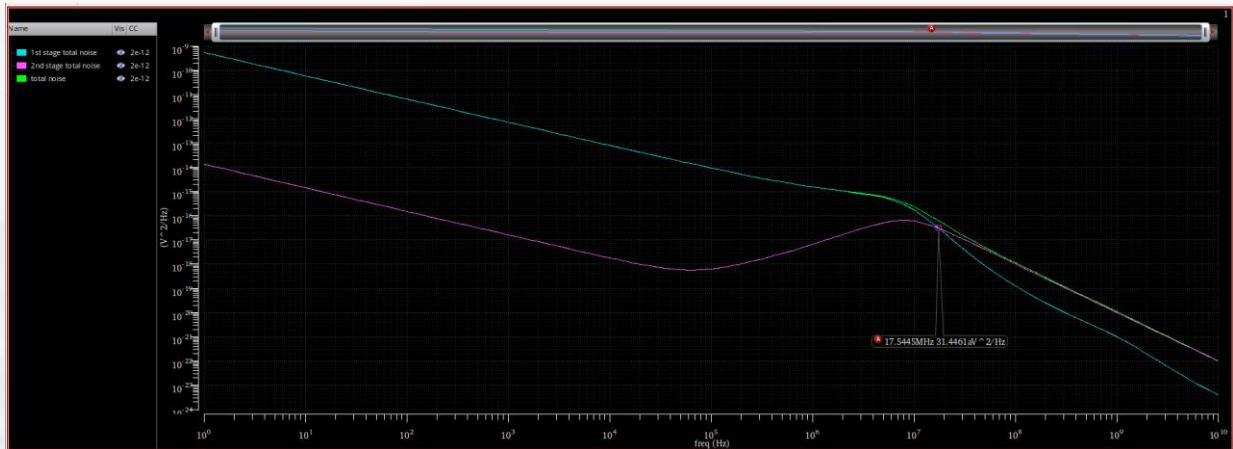


1st stage noise: ■ 2nd stage noise ■ 1st stage noise ■ total noise $f_c \approx 1.86\text{MHz}$



2nd stage noise: ■ thermal noise ■ flicker noise ■ total noise $f_c \approx 2.38\text{MHz}$

The above graphs show that the dominant noise term is the flicker noise at low frequency, and then the thermal noise dominates at high noise.



OTA total noise: ■ 2nd stage noise ■ 1st stage noise ■ total noise $f_c \approx 17.54\text{MHz}$

The above graph show that the 1st stage noise is the dominant term in the total noise, but at high frequency the 2nd stage dominate over the 1st stage duo to the degradation in the 1st stage MOSFETs because of the parasitic capacitance.